

Lab Assignment :6 Information Security

Deadline:11:59pm 11/19/25

Note: Add inlab activities with output in lab report and Submit lab report in pdf .

Task 1: Rail Fence Cipher :

Instructions:

1. Take the plaintext as:
2. <YourFullName> <space> <YourFavouriteCourse>

(Example: "Ayesha Khan DataScience")

3. The **number of rails (key)** = number of characters in **your first name**.
(If first name is "Ayesha", key = 6)
4. **Perform the following manually (on paper or document):**
 - Write the plaintext in a zigzag pattern according to the number of rails.
 - Read row-by-row to form ciphertext.
 - Show the exact zigzag pattern in your dry run.
 - Perform **decryption** step-by-step to recover the original message.
5. After manual encryption and decryption, run the same input using the **Rail Fence Cipher code provided in lab**.
6. Compare your **manual** and **code** outputs and state whether both match.